

Lecture 13

Outliers, Normal Quantiles, Transformations and the Margin of Error

Review

1 What is a density curve?

A smooth curve that describes a distribution; the total area under it is 1.

2 What is the standard normal distribution?

The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$.

3 What is a z-score?

The number of standard deviations a value falls from the mean: $z = (x - \bar{x}) / s$.

4 What is the empirical rule?

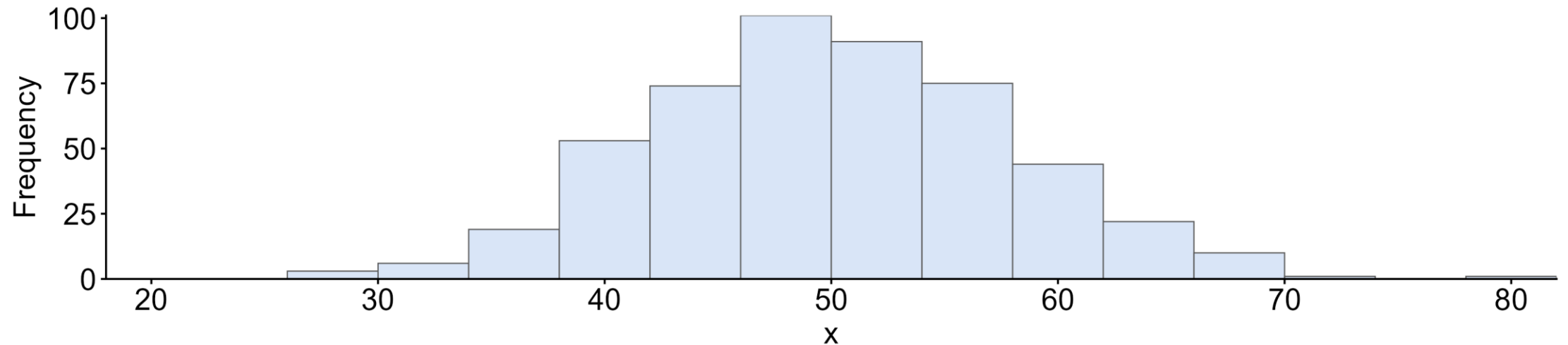
About 68%, 95% and 99.7% of values fall within 1, 2 and 3 SDs of the mean (bell-shaped only).

Review: from x to z

A variable x has the distribution shown below.

$\bar{x} = 50$ $s = 8$

How many standard deviations from its mean is an observation of $x = 62$?



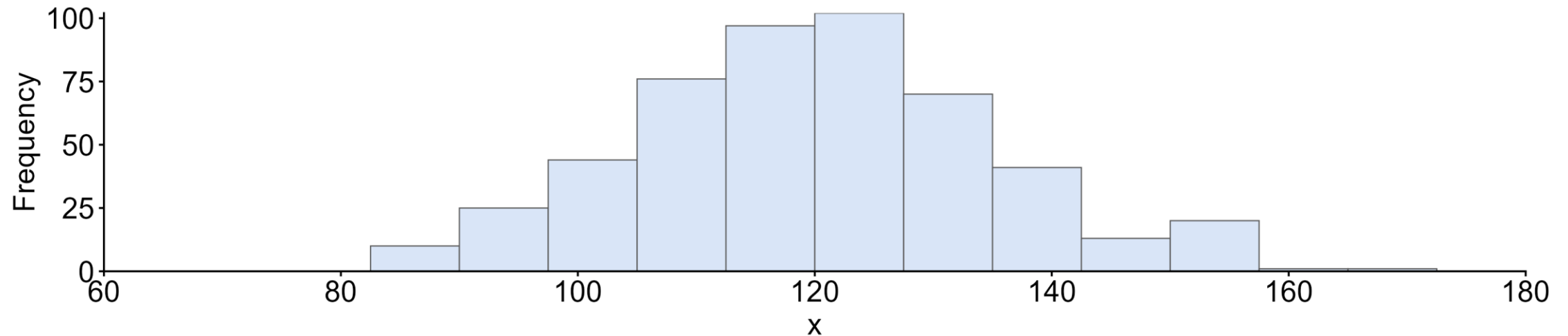
$z = (62 - 50) / 8 = 1.5$: 1.5 standard deviations above the mean

Review: from z to x

A variable x has the distribution shown below.

$\bar{x} = 120$ $s = 15$

What value of x marks 3 standard deviations above the mean?



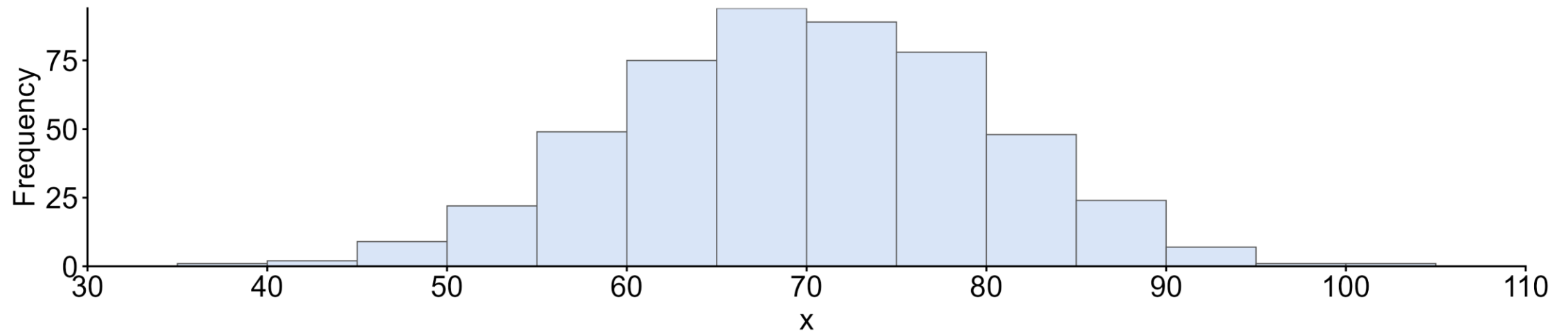
$$x = \bar{x} + z s = 120 + 3(15) = 165$$

Review: the empirical rule

A variable x has the distribution shown below.

$\bar{x} = 70$ $s = 10$

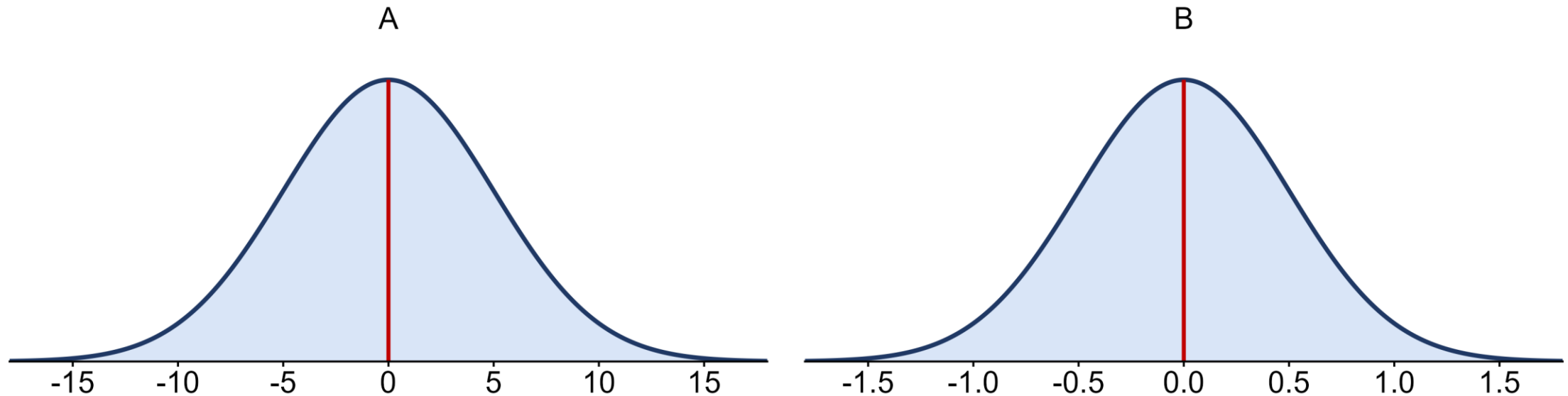
Approximately what value of x is the 16th percentile?



$50\% - 34\% = 16\%$ of values lie below $\bar{x} - s$, so $x \approx 70 - 10 = 60$

What is the standard deviation?

Estimate σ for curve A and for curve B.



A: $\sigma \approx 5$

B: $\sigma \approx 0.5$

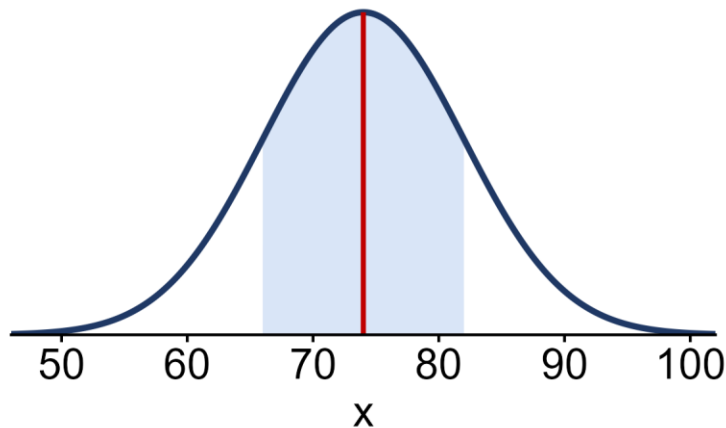
Example: using the empirical rule

A variable x has the following distribution.

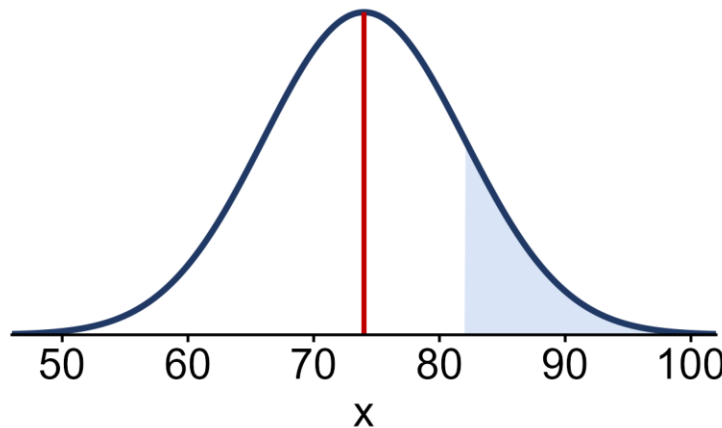
$$\bar{x} = 74 \quad s = 8$$

What percentage of the values of x falls in each shaded region?

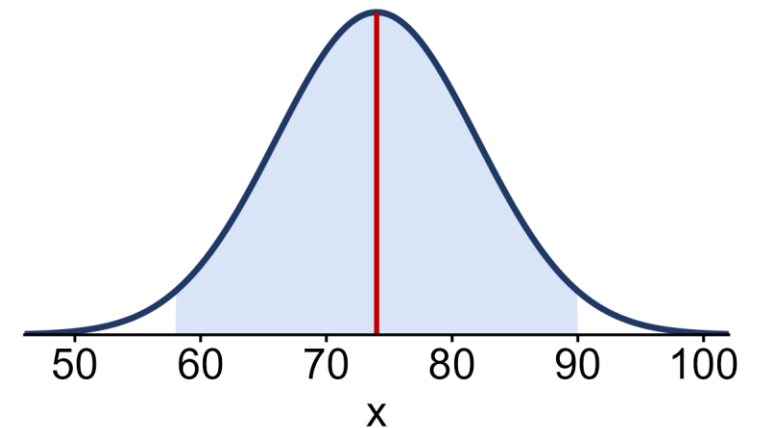
(a) 66 to 82



(b) 82 and above



(c) 58 to 90



(a) about 68%

(b) about 16%

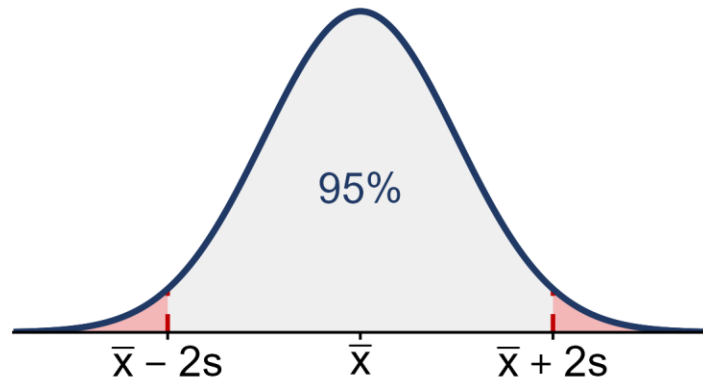
(c) about 95%

Outliers: the two-standard-deviation rule

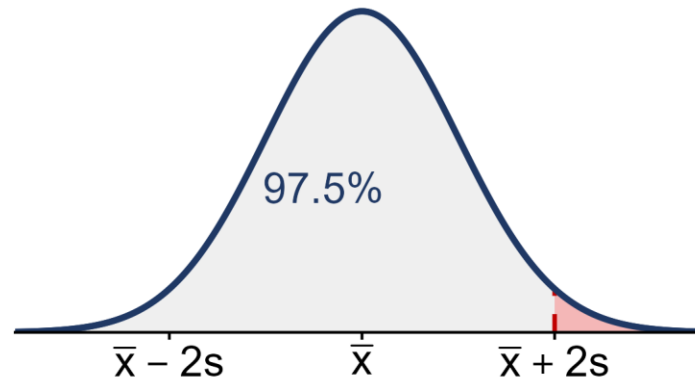
outlier

a value at least 2 standard deviations from the mean, $|z| \geq 2$ (about 5% of bell-shaped data)

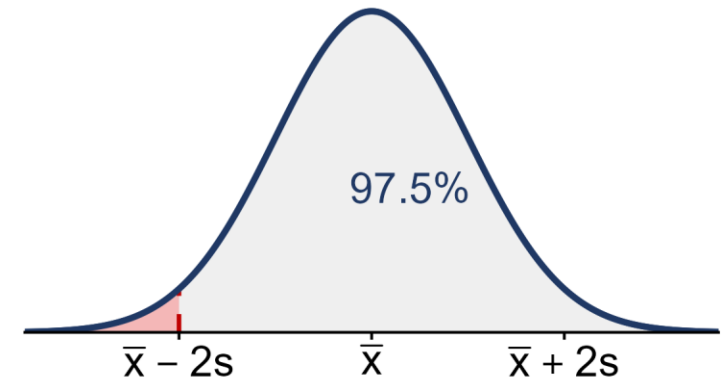
both tails: $2.5\% + 2.5\% = 5\%$



upper tail only: 2.5%



lower tail only: 2.5%



Data: student heights

A survey of 379 college students recorded each student's height and gender.

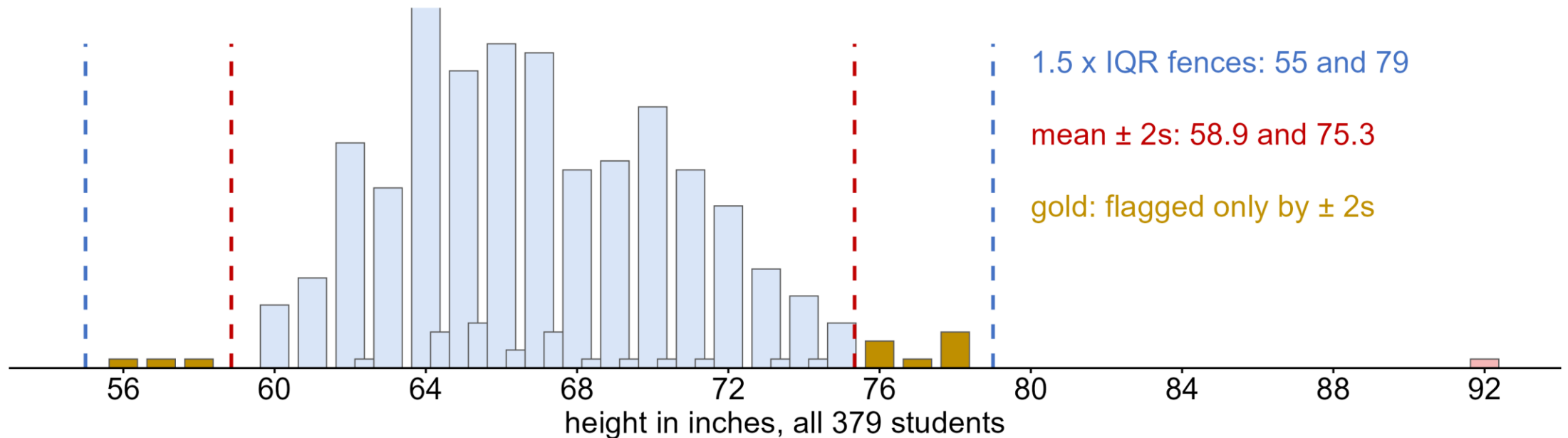
X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
...	...

n = 379 students (the file lists the women first, then the men, each sorted by height)

When the two outlier rules disagree

Do the $1.5 \times \text{IQR}$ rule and the $\pm 2s$ rule flag the same students?



No: the IQR rule flags only 1 student (92 in), the $\pm 2s$ rule flags 12
Why: $\pm 2s$ flags about 5% of any bell-shaped data; $1.5 \times \text{IQR}$ flags under 1%

Computing quantiles from a normal distribution

Heights of all 379 students are roughly bell shaped.

$\bar{x} = 67.1$ in $s = 4.12$ in one student is 72 in

- (a) What is the student's z-score?
- (b) What percentage of the students are shorter?

- (a) $z = 1.19$
- (b) The empirical rule cannot say: 1.19 is not a whole number of standard deviations

How it used to be done: the Z-table

the standard normal table

each entry is the area to the LEFT of z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545

$z = 1.19$: row 1.1, column .09 → 0.8830, so about 88% are shorter

Today: the computer does the lookup

From a value to an area: pnorm

pnorm(1.19) = 0.8830

From an area to a value: qnorm

qnorm(0.90) = 1.28

No R in front of you?

The normal calculator at stat-tools.jmks-homelab.dev does both.

A note about transformations of variables

Changing units

We often need to change the units of a variable: °F to °C, feet to meters, dollars to euros.

Linear transformations: adding, subtracting, multiplying, dividing

$y = ax + b$ (a scales, b shifts). The z-score is a linear transformation. They preserve the shape of the distribution.

Nonlinear transformations: squaring, taking roots, logarithms, exponentiation

They do NOT preserve the shape of the distribution.

What a linear transformation does to the summaries

The center moves with the whole transformation

$\bar{y} = a\bar{x} + b$, and the same for the median.

The spread ignores the shift

s and IQR are multiplied by $|a|$; b has no effect.

The shape does not change at all

Skewness, gaps and outliers all survive unchanged.

Example: inches to centimeters

The 379 heights are converted from inches to centimeters: $y = 2.54x$.

$\bar{x} = 67.1$ in $s = 4.12$ in the 72-inch student = 182.9 cm

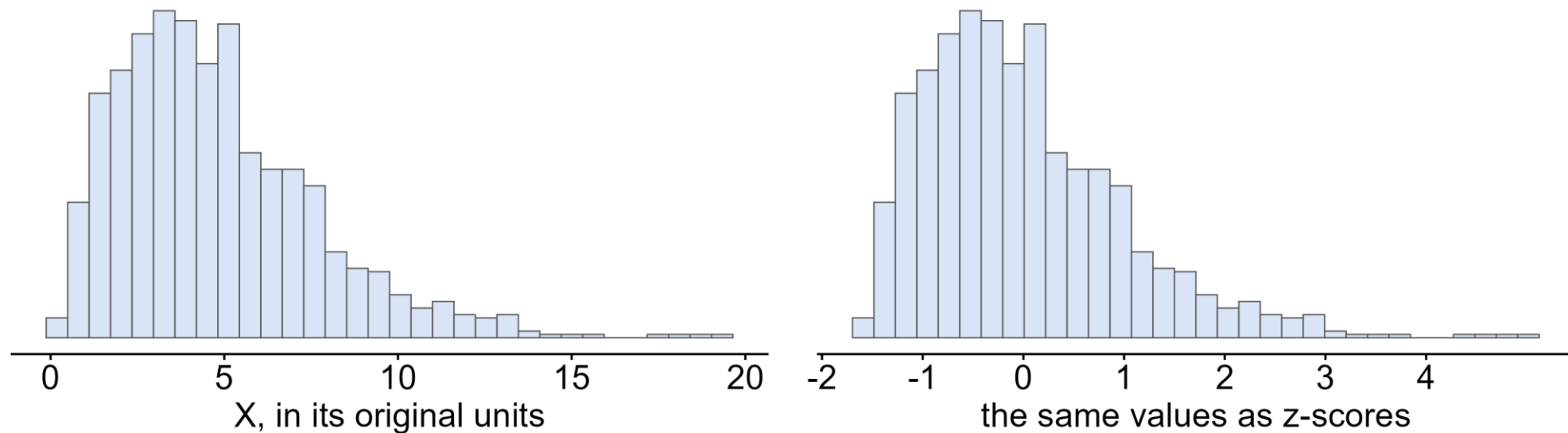
- (a) What are the mean and standard deviation in centimeters?
- (b) What is the student's z-score in centimeters?

(a) $\bar{y} = 170.4$ cm, $s = 10.46$ cm

(b) $z = 1.19$, unchanged

Does a z-score make data normal?

Does converting skewed data to z-scores make it normal?

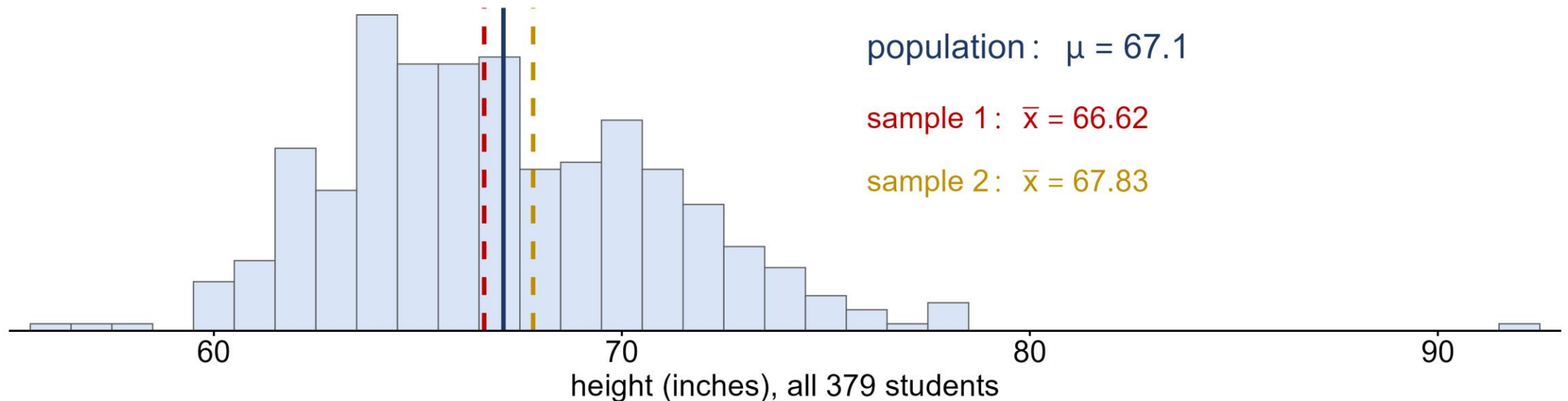


No: standardizing changes the axis, not the shape

Estimating a mean from a sample

Treat all 379 students as the population, and estimate μ from a random sample of $n = 20$.

Why do the two samples miss μ , and by different amounts?

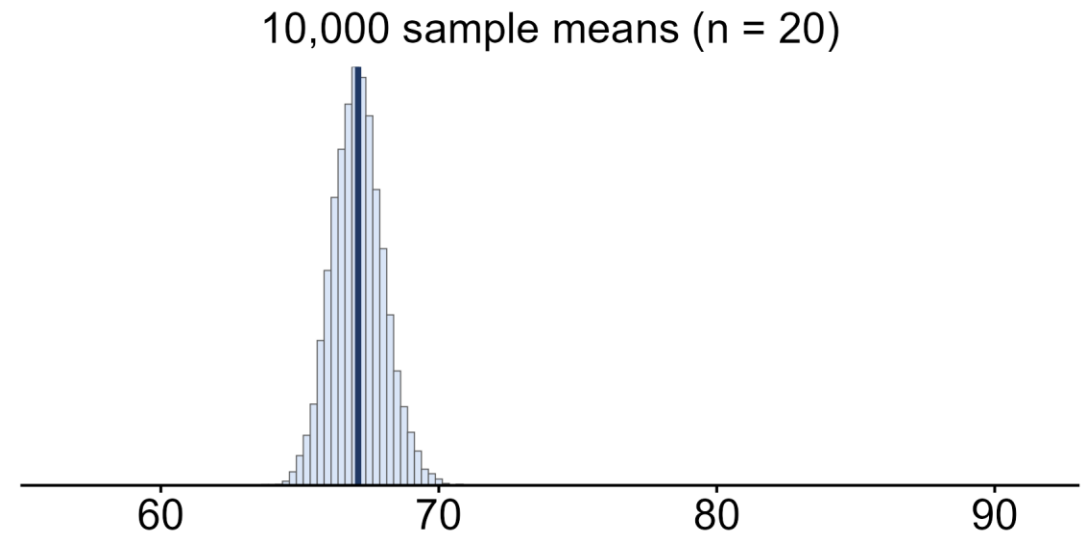
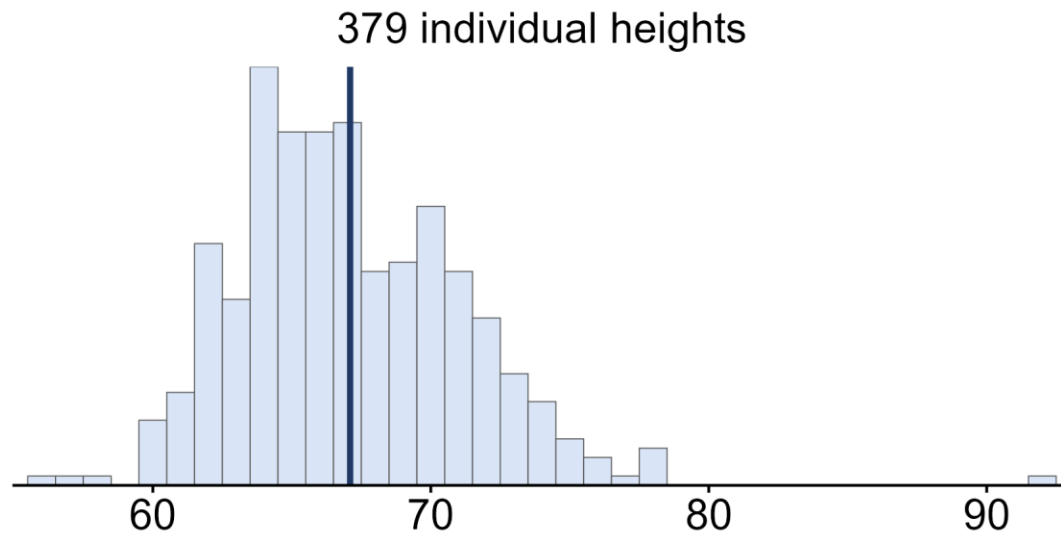


Each sample holds different students, so each $\bar{x}$ misses by a different amount: sampling error

Sampling distributions

sampling distribution

the distribution of a statistic, such as $\bar{x}$, over all possible samples of the same size n



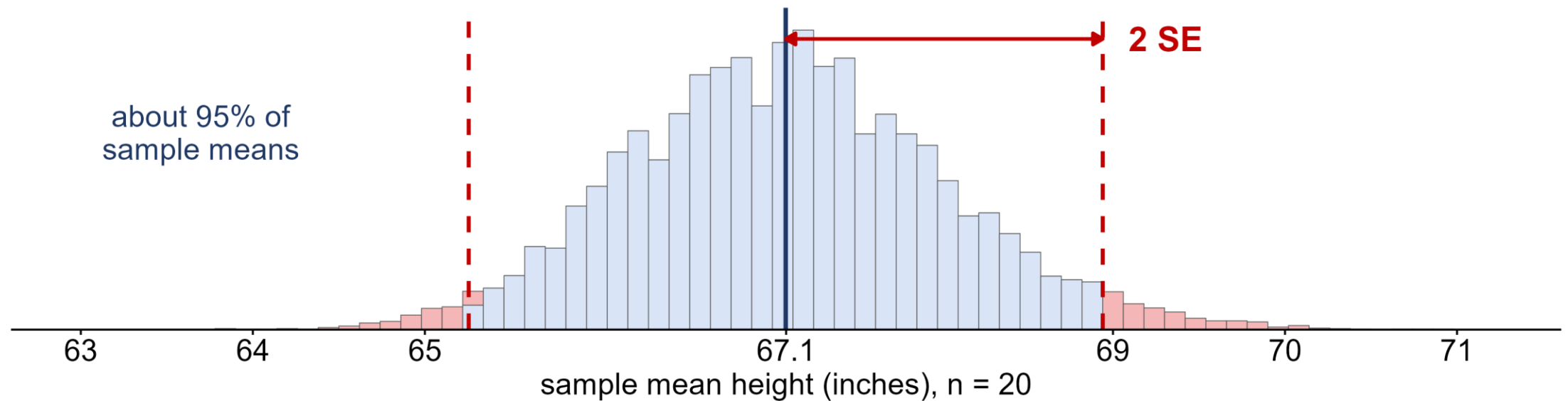
You've seen these before:

$SE = s / \sqrt{n}$ (Week 3) is its spread; the Rec Center histograms (Week 4) were sampling distributions

Margin of error

margin of error

how far we expect an estimate to fall from the parameter: the largest distance to an estimate that is not an outlier, $2 \times \text{SE}$



Example: the margin of error for $\bar{x}$

Random samples of $n = 20$ from all 379 students.

$\sigma = 4.12$ in $n = 20$

What are the standard error and the margin of error of $\bar{x}$?

$SE = \sigma / \sqrt{n} = 4.12 / \sqrt{20} \approx 0.92$ in

Margin of error = $2 \times 0.92 \approx 1.8$ in: about 95% of samples land within 67.1 ± 1.8 in